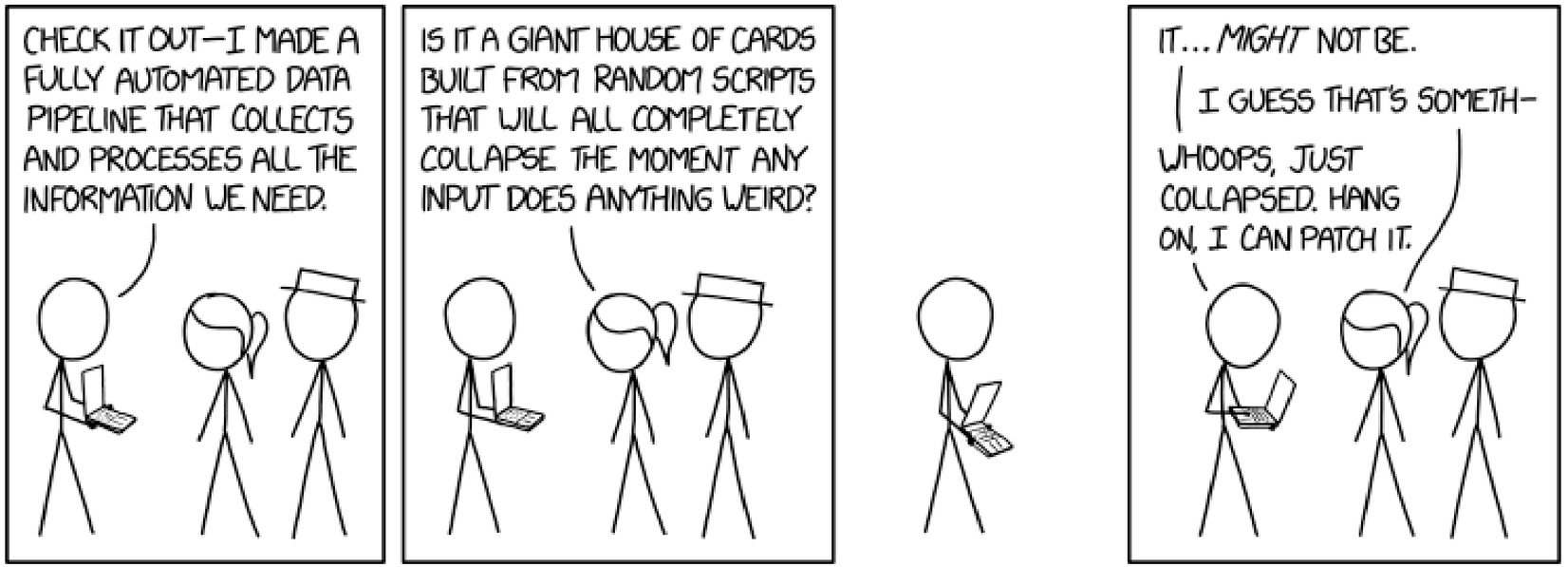
universität innsbruck
Department of Computer Science

Organization

Data Engineering and Analytics

Eva Zangerle



Comic taken from https://xkcd.com/2054/

Goals

Students who complete this course successfully will be able to:

- Perform data engineering and analyses tasks.
- Understand and apply machine learning and data analytics techniques for the systematic analyses of large datasets.
- Identify, devise and evaluate meaningful features for analyses and modeling.
- Evaluate and compare data engineering and analysis techniques for a given use case.
- Carry out reproducible, documented, and efficient data analyses and communicate the results.
- Create datasets (including gathering, generation, cleaning, and validation thereof).
- Understand the limitations, weaknesses, and potential biases of data collection and analysis methods.

Syllabus

- Introduction
- Data preparation and quality
- Data set analyses (exploratory, descriptive, visualization)
- Hypotheses, research questions, assumptions
- Statistical tests
- Modeling and prediction
- Reproducible research
- Biases

Course Material

- Slides (OLAT)
- Current blogs and papers on relevant topics (slides/OLAT)
- Jupyter Notebooks (UIBK gitlab)

Course Material (see OLAT)

- Data Science from Scratch: First Principles with Python; Joel Grus; O'Reilly UK Ltd.; 2nd edition, 2019
- Fundamentals of Data Engineering; Joe Reis, Matt Housley; O'Reilly UK Ltd.; 1st edition, 2022
- Practical Statistics for Data Scientists: 50+ Essential Concepts Using R and Python; Peter Bruce, Andrew Bruce, Peter Gedeck; O'Reilly UK Ltd.; 2nd edition, 2020
- Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython; Wes McKinney; O'Reilly UK Ltd.; 3rdnd edition, 2022
- Think Like a Data Scientist: Tackle the data science process step-by-step ; Brian Godsey; Manning Publications; 1st Edition, 2017
- Cleaning Data for Effective Data Science: Doing the other 80% of the work with Python, R, and command-line tools; David Mertz; Packt Publishing, 2021
- Experimental design and analysis; Howard J. Seltman; Carnegie Mellon University, 2018; available online
- Downey, Allen B. Think stats. O'Reilly Media, Inc., 2011; available online
- Duboue, Pablo. The Art of Feature Engineering: Essentials for Machine Learning. Cambridge University Press, 2020.
- Zheng, Alice, and Amanda Casari. Feature engineering for machine learning: principles and techniques for data scientists. O'Reilly Media, Inc., 2018.

Code

- Python (Jupyter notebooks)
- Diverse selection of datasets to be used in lecture and tutorial.
- Notebooks used during lecture available on Gitlab.
- Please use issue issue tracker to report issues and contribute via merge requests.

Code

- Small snippets on slides
- Sources to examples in lecture are available via Jupyterhub, indicated by a link to the notebook and section in the footer.

```
from math import nan, pi
for n in range(7, 10):
exp1 = 2\*\*n
a = (22/7) \*\* exp1
b = pi \*\* exp1
```

Grading

- Lecture exam 1: January 27, 2027, 08:30-10:30, HSB3
- Lecture exam 2: April 15, 2027, 09:00-11:00
- Lecture exam 3: September 30, 2027, 09:00-11:00
- Pen and paper exam, 90 minutes

Grading

- Exam registration starts five weeks before the exam.
- Exam registration ends two weeks before the exam (no late registration via email).
- De-registration possible until 72 hours before the exam.
- If you fail to appear without providing valid reason, you are not permitted to take the examination at the next examination date (see guidelines).

Tutorial

- Deepen understanding of lecture contents.
- Practical application of lecture contents.
- Coding of data analyses.
- **Discussion and presentation of findings.**
- **Interpretation of results.**
- Applicability of methods.
- First tutorial: October 14, 2026 (Sheet 1 due October 13, 2026)
  - 10:15 (group Zangerle, RR19)
  - 12:00 (group Gassler, 3W04)

Survey

Your prior experiences and knowledge?

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2586 6246

My research

- Recommender Systems (specialization course next semester!)
- User Modeling
- Music Information Retrieval
- Sometimes also Natural Language Processing, Databases and Indexing

Showcase Study

Kowald et al. *EPJ Data Science* (2021) 10:14
https://doi.org/10.1140/epjds/s13688-021-00268-9

EPJ .org

EPJ Data Science
a SpringerOpen Journal

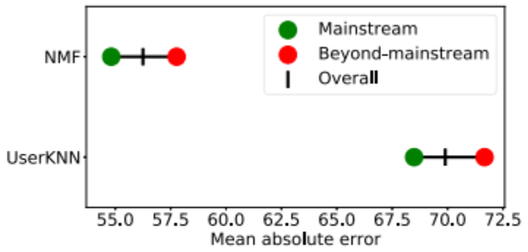
REGULAR ARTICLE Open Access

Support the underground: characteristics of beyond-mainstream music listeners

Dominik Kowald[1*], Peter Muellner[1], Eva Zangerle[2], Christine Bauer[3], Markus Schedl[4,5] and Elisabeth Lex[6*]

Showcase Study (continued)

Figure 1 Recommendation accuracy measured by the mean absolute error (MAE) of a non-negative matrix factorization-based approach (i.e., NMF [10]) and a neighborhood-based approach (i.e., UserKNN [11]) for mainstream and beyond-mainstream user groups in Last.fm. We see that beyond-mainstream users receive a substantially lower recommendation quality (i.e., higher MAE) compared to mainstream music listeners. Thus, for recommender systems, it is harder to provide high-quality recommendations to beyond-mainstream than to mainstream listeners



Showcase Study (continued)

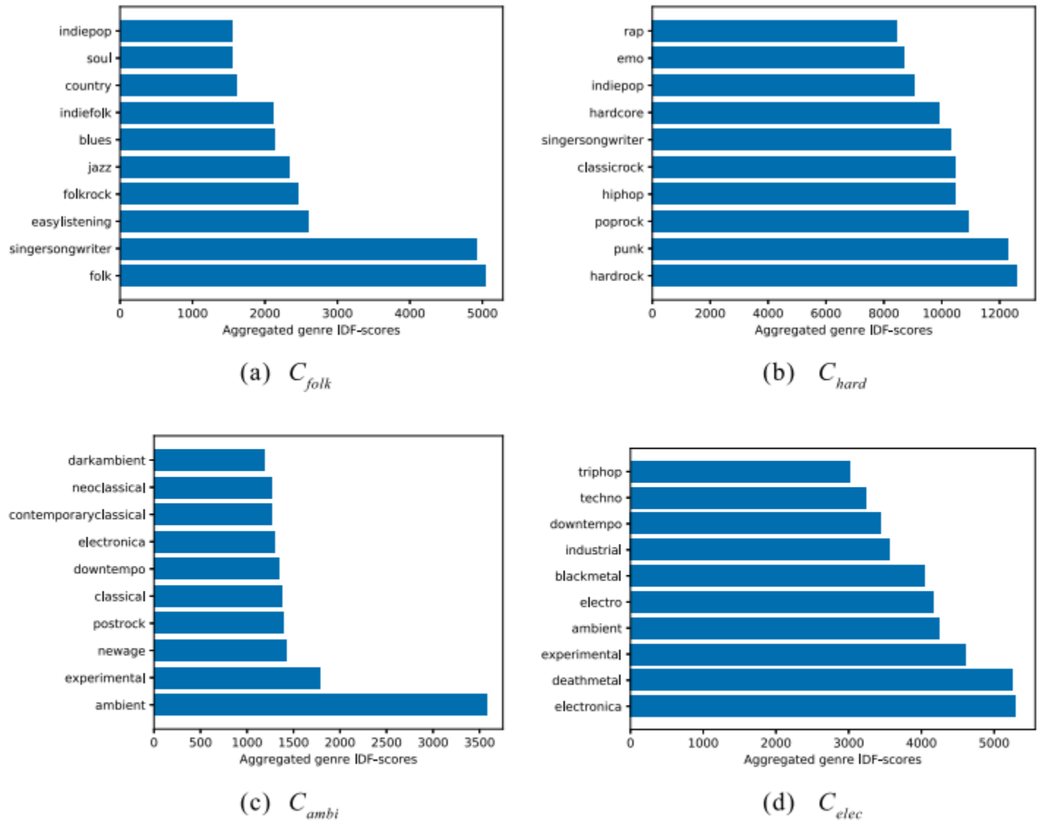


Figure 7 Top-10 genres of the four music clusters C_1–C_4 according to the aggregated genre IDF-scores. We name the clusters according to the top genre, i.e., (**a**) blue (hatch: /) → C_{folk} ("folk"), (**b**) green (hatch: +) → C_{hard} ("hardrock"), (**c**) orange (hatch: o) → C_{ambi} ("ambient"), and (**d**) pink (hatch: x) → C_{elec} ("electronica")

Showcase Study (continued)

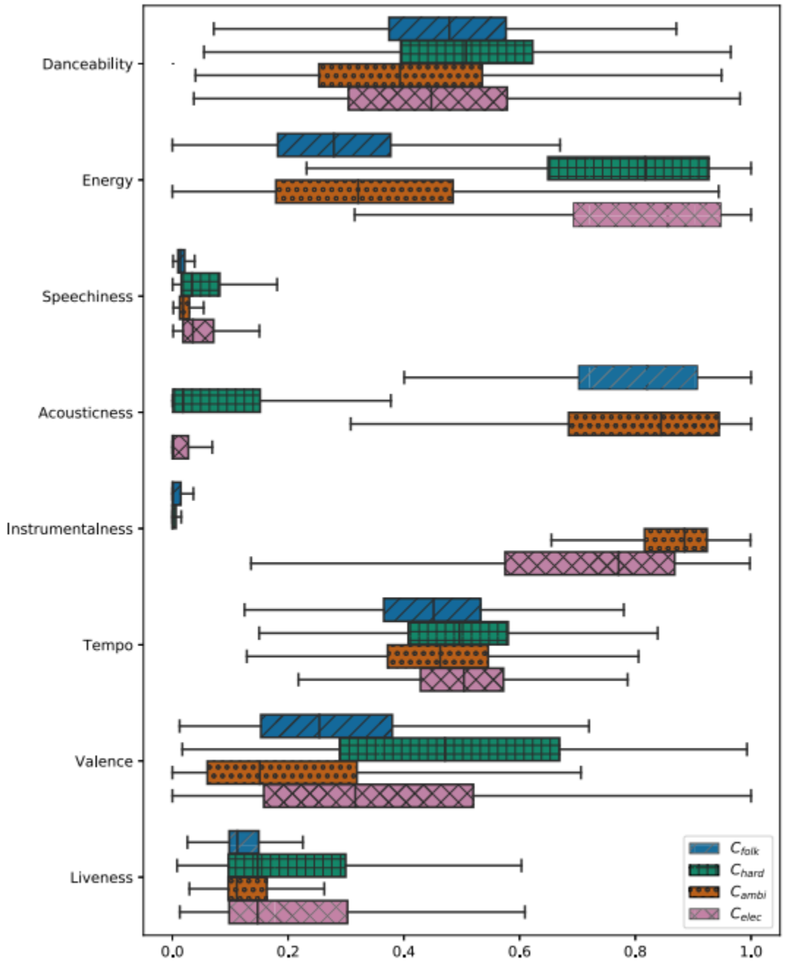
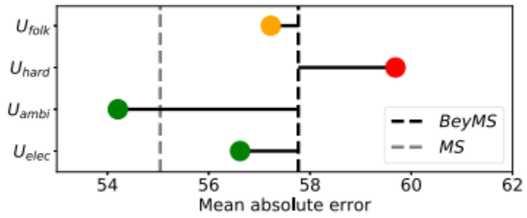


Figure 9 Distribution of the eight acoustic features for the four music clusters. While the clusters do not show obvious differences with respect to danceability and tempo, we find large differences with respect to energy, acousticness and instrumentalness

Showcase Study (continued)

Figure 13 Comparison of the mean absolute error (MAE) scores reached by NMF for the four subgroups with the ones reached by NMF for *BeyMS* (black dashed line) and *MS* (gray dashed line). While specific subgroups (i.e., U_{hard}) are treated in an unfair way by recommendation algorithms, others (i.e., U_{ambi}) are not



Questions, Feedback, ...

- Before, during, after lecture and tutorial
- Email: eva.zangerle@uibk.ac.at
- Virtual office hours: Tuesday, 15:00-16:00 – please book in advance